



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

COMPUTER SCIENCE
PAPER 1

6023/1
3 hours

SPECIMEN PAPER

Additional materials:
Answer booklet
Scientific calculator [Non-programmable]

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number provided on the answer paper/answer booklet.

Answer **all** questions.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This specimen paper consists of 4 printed pages.

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- 1
 - (a)
 - (i) Define the term *underflow* with respect to binary representation. [2]
 - (ii) Discuss the impact of underflow on precision. [2]
 - (b) Compare the efficiency of ASCII and Unicode as data representation schemes for storing a single character. [2]
 - (c)
 - (i) Convert the decimal number 109 to 8-bit binary. [2]
 - (ii) Assess the implication of using fixed-width representation for computing and data processing. [4]

- 2 Normalise the following floating point numbers.
 - (i) 0.0001000 0001₂ [2]
 - (ii) 11011.101 1001₂ [2]

NB the bits for mantisa is 8 and bits for the exponent is 4.

- 3
 - (a) For the IP address 192.100.18.33:
 - (i) identify the class IP address, [1]
 - (ii) state the subnet mask of the IP address, [1]
 - (iii) convert the IP address in (ii) to binary format. [2]
 - (b) For the following URL
[http: // www.zimsec.co.zw / Exams / computer science. Pdf](http://www.zimsec.co.zw/Exams/computer%20science.Pdf)
 identify all the parts of the URL. [4]
 - (c) Evaluate the role of Domain Name System (DNS) in network addressing. [2]

- 4 NOCZIM has benefited from cloud computing for data backup.
 - (a) Describe **two** cloud service models. [6]
 - (b) Explain **two** disadvantages of cloud computing. [4]

- 5 (a) Explain how the separation of instruction and data memories on Harvard Architecture influences system performance. [4]
- (b) Describe the stages of the fetch / decode / execute cycle, paying attention to how the special registers in the processor are used.
- NB:** Exclude a jump statement in your answer. [7]
- (c) Assess the role of any **three** buses used in Von Neumann model in facilitating the fetch-execute cycle and overall system efficiency. [6]
- 6 Design a logic circuit for the following logical expression. [3]
- $$(A + B). \bar{C}$$
- 7 (a) Define the term data *integrity*. [1]
- (b) A doctor plans to share patients' records on a computer system.
- (i) State **three** data protection principles that the doctor should obey. [3]
- (ii) Explain **two** possible effects if the doctor does not obey the data protection principles. [4]
- (iii) Suggest **one** biometric technique to protect data. Explain how it works. [2]
- 8 (a) Design a flowchart showing steps to be followed when seeking admission to a new school. Use the following phrases. [10]
- Search for school, Prepare for the admission, Did you pass the exam, Submit necessary documents, End, Are places available, Start, Is there an admission test.
- (b) The following is a list of elements that need to be sorted.
- 3 5 2 7 4 1
- Demonstrate how the elements can be sorted using a bubble sort algorithm. Show output for each pass. [5]

- 9 (a) Explain the term *data redundancy* giving a suitable characteristic which leads to it. [2]
- (b) Explain the following SQL command.
- CREATE TABLE Students (

id INT PRIMARY KEY
name VARCHAR (255)); [7]
- 10 (a) Define the term *financial literacy*. [1]
- (b) You are the owner of an online shop. Customers pay through money transfers using credit cards, ZIPIT or mobile wallet.
- Assume that you can select only one of these methods of payment.
Which one would you select?
Give four reasons to justify your choice. [4]
- (c) Computer Science plays a very important role in the field of engineering.
- Explain **five** ways how Computer Science can help engineers in the simulation of prototype designing. [5]